



Cortexa

Architecture and execution

Current implementation · 7 October 2026

Source `bc12776412c717613de1fdc42c38bf3312493726`

Merged main `bbe7546281fec3c8b4b608d68a52bc9732d9ecd9`

Nine connected views. Solid arrows carry labeled data; dashed lines mark separate foundations or unimplemented directions. Read the companion HTML for full capability, security and adoption evidence.

Native acceptance covers the recorded API/Codex variants. Other implementations, fixtures and future work are explicitly distinguished. No product tests, builds or provider calls were repeated for this atlas.

Ledger33 unchanged · ECC hooks/MCP disabled · D-125/M1/M2 parked.

This documentation is not a complete security audit.

01 · System overview

Local desktop coordination, explicit provider boundaries, no general device executor

Native

Interface

Owner

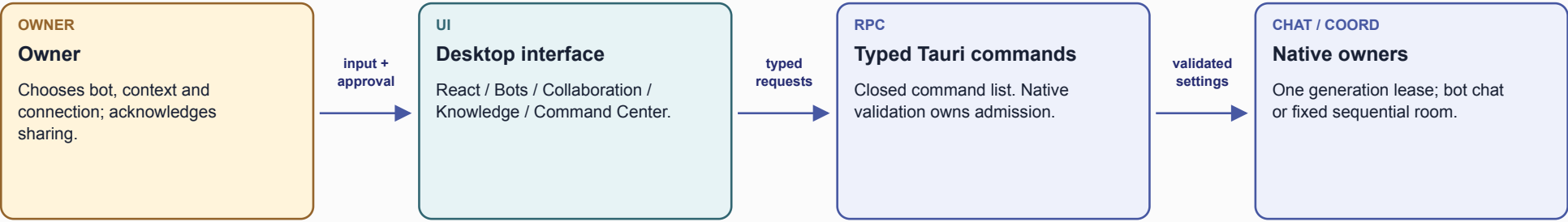
Remote

Stored

Separate foundation

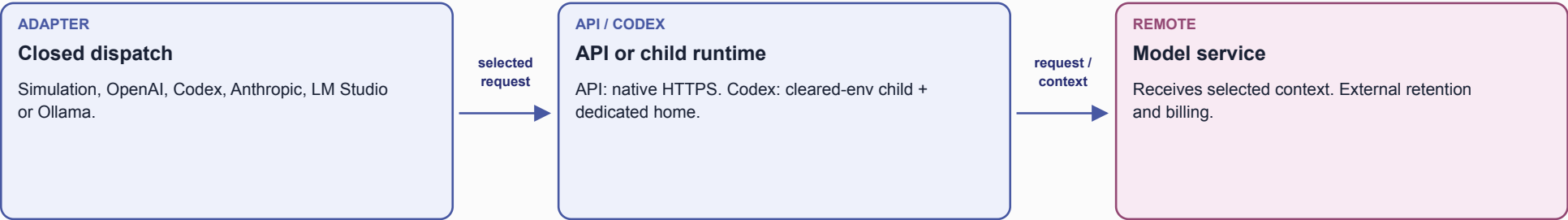
Deferred / unverified

Interactive request · WebView data is untrusted



Evidence: UI, RPC, CHAT, COORD

Outbound alternatives · local transport crosses into an external service



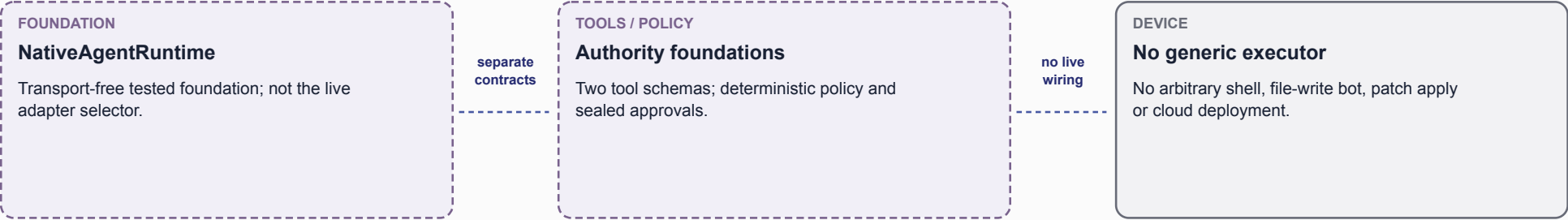
Evidence: ADAPTER, API, CODEX, CLAUDE, LOCAL

Explicit persistence · selected content, not automatic learning



Evidence: PROFILE_DB, KNOW_DB, DB, ROOM_DB, DIAG

Execution boundary · foundations are not a live tools pipeline



Evidence: FOUNDATION, TOOLS, POLICY, APPROVAL

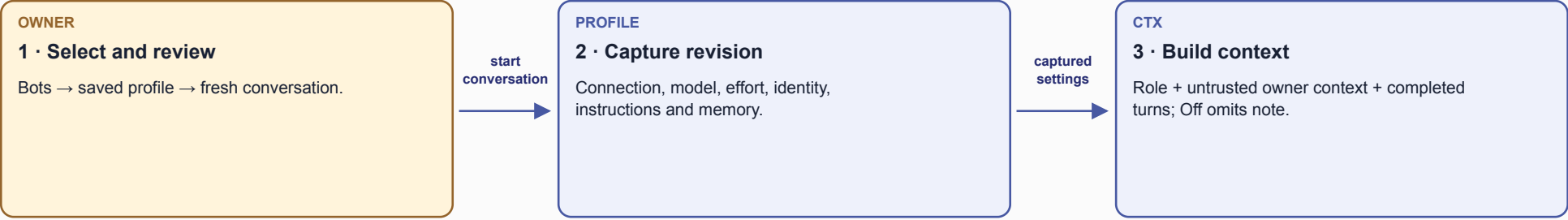
Rows show distinct paths, not one serial workflow. Anthropic/local adapters have code and offline coverage; the accepted live matrix is OpenAI/Codex. Legacy Conversations includes mocks and a fixed native sample.

02 · Prompt to completion

Single-bot native path; numbered rows continue in order

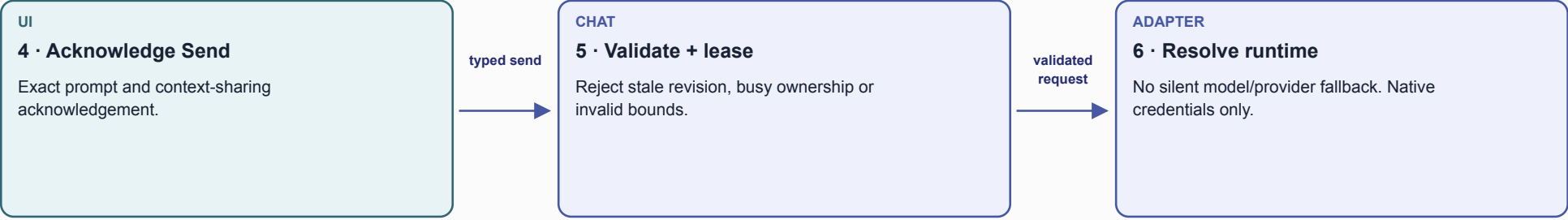
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

1–3 · Before generation



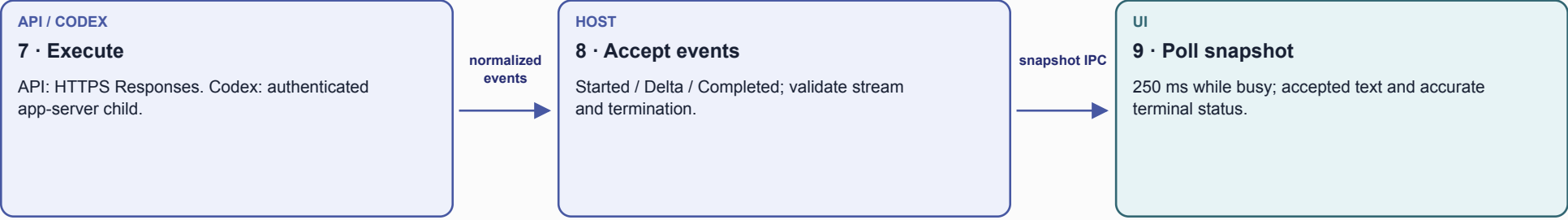
Evidence: UI, PROFILE, CTX

4–6 · Native admission



Evidence: UI, CHAT, ADAPTER

7–9 · Selected runtime only



Evidence: API, CODEX, HOST, CHAT, UI

10–12 · Settlement



Evidence: CHAT, CTX, DIAG

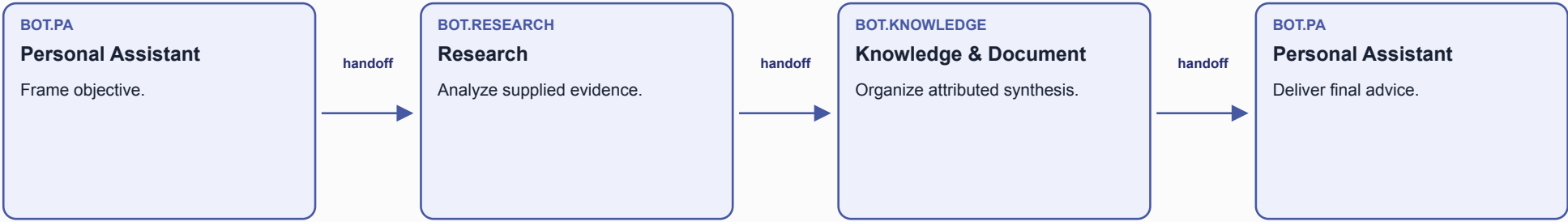
No general tool-action stage exists here. Simulation needs no remote authentication. Collaboration instead persists room snapshots. Partial text alone does not prove completion.

03 · Collaboration routes and delivery

COORD is deterministic application coordination; every live route is sequential

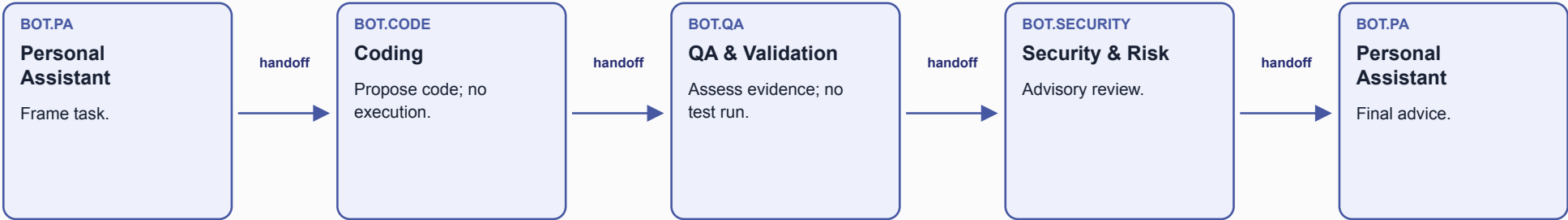
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Research · four model stages · retained minimum live route



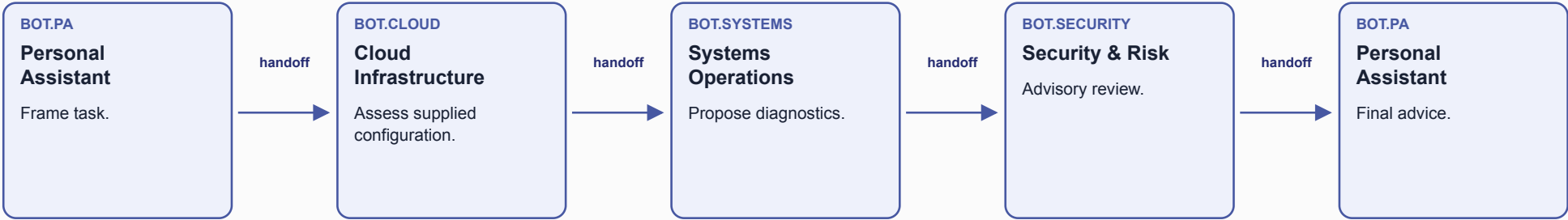
Evidence: ROUTES, COORD

Engineering · five stages · other live variant not accepted by Research evidence



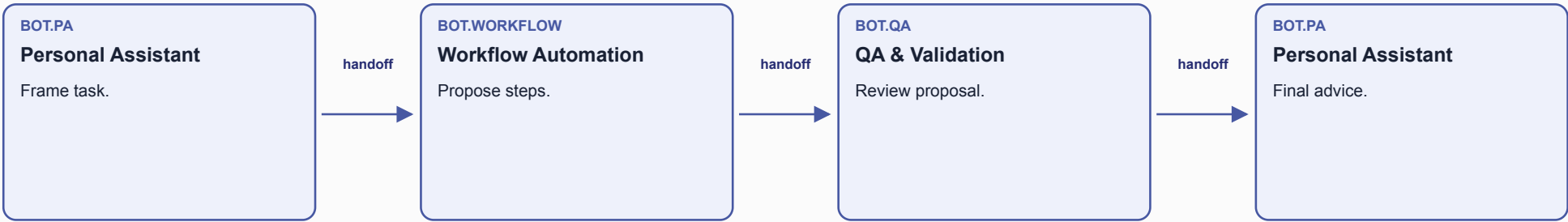
Evidence: ROUTES, CATALOG

Operations · five stages · advice, not machine administration



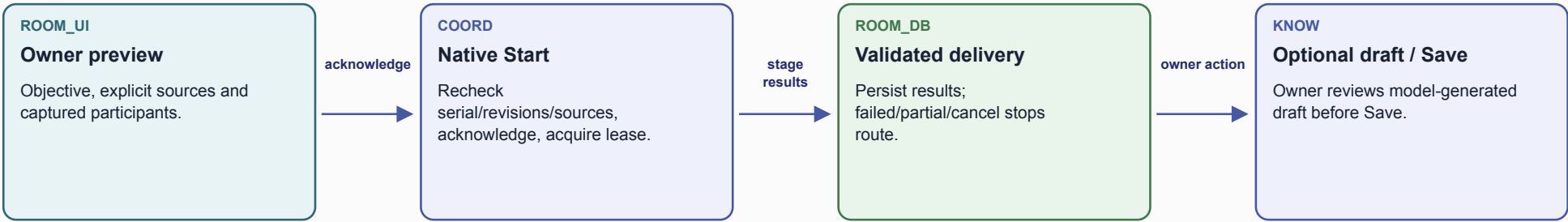
Evidence: ROUTES, CATALOG

Proposal · four stages · no autonomous scheduler



Evidence: ROUTES, CATALOG

Common envelope · sharing approval is not patch-apply authority



Evidence: COORD, ROOM_DB, KNOW

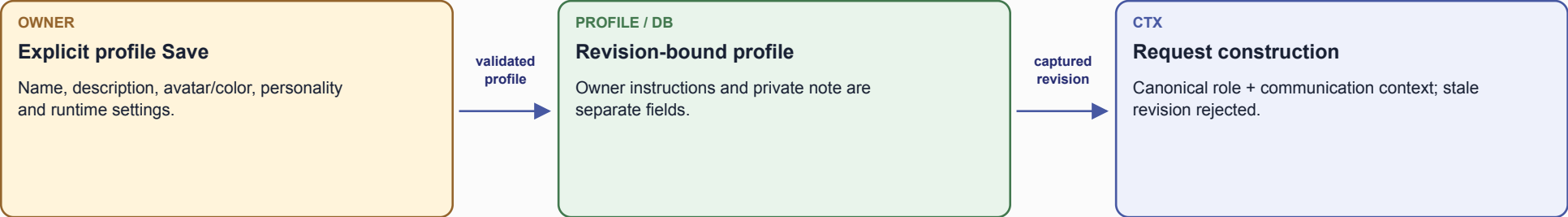
No parallel model agents in these routes. Handoff validation is not factual verification. A Security bot cannot override native policy. Foundation parallelism is separate.

04 · Identity, instructions and memory

Separate stores and sharing rules; none grants device authority

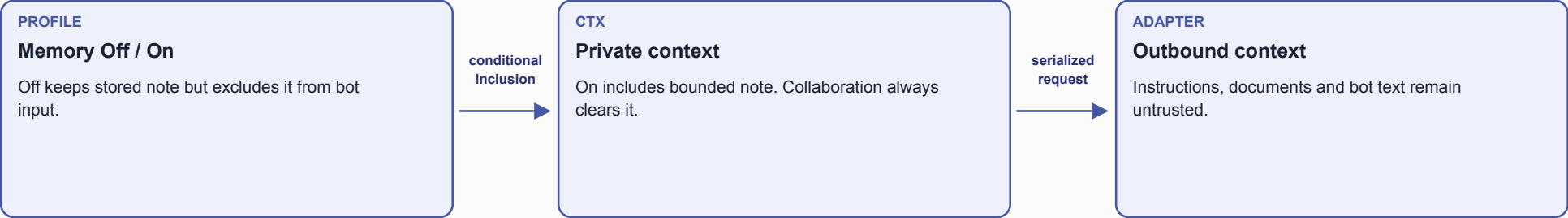
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Saved personality and settings



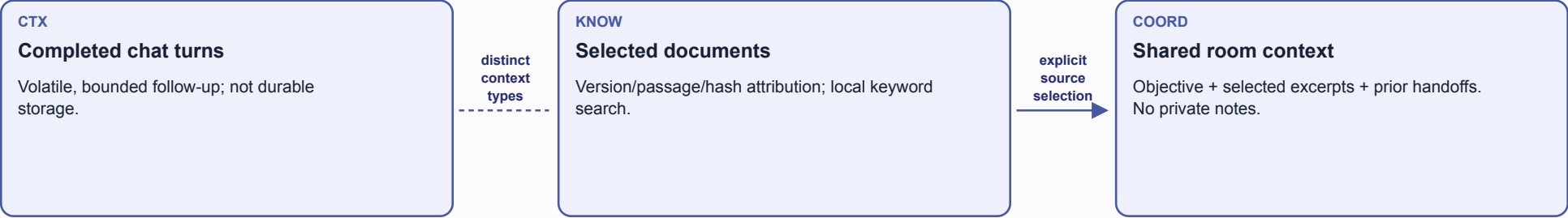
Evidence: PROFILE, PROFILE_DB, CTX

Private note · optional prompt context, not automatic learning



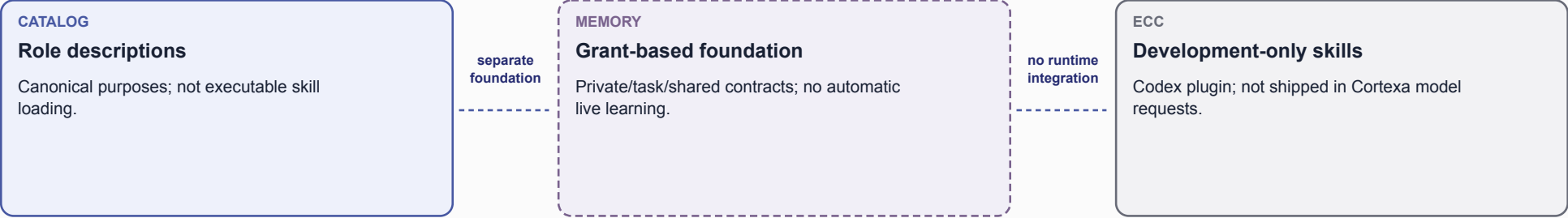
Evidence: PROFILE, CTX, ADAPTER

History versus shared Knowledge



Evidence: CTX, KNOW_DB, COORD

Skills and future memory are separate



Evidence: CATALOG, MEMORY, PLAN

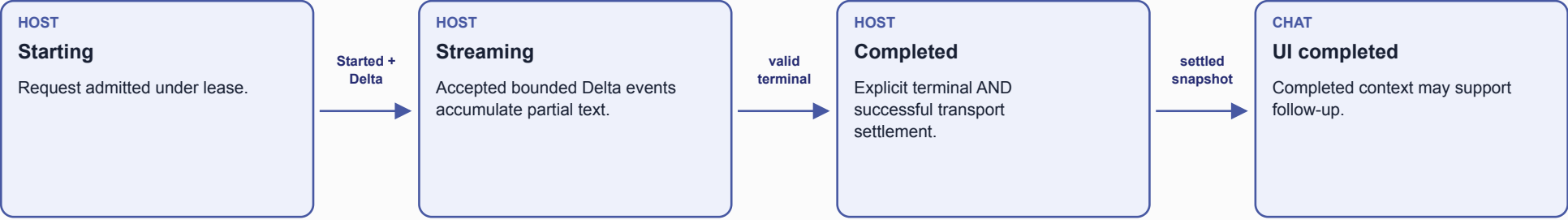
Knowledge provides Markdown, versions, stable links, bounded graph and owner import/export. No embedding retrieval, external vault sync or automatic private-memory learning is established.

05 · Lifecycle and failure handling

Exact state vocabularies differ between host, bot UI and persisted rooms

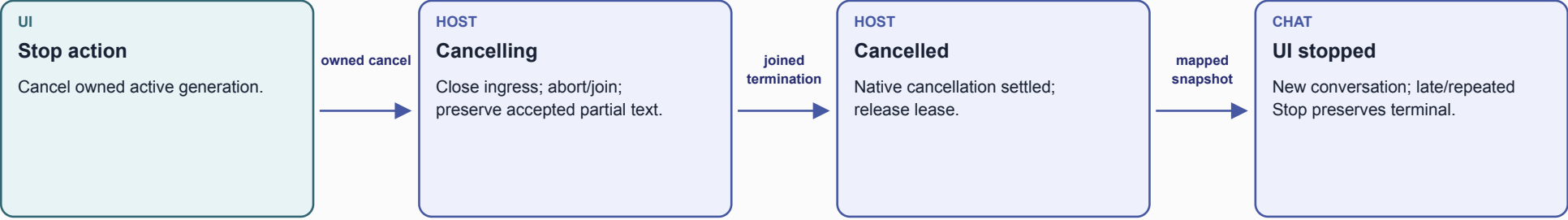
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

HOST · successful generation



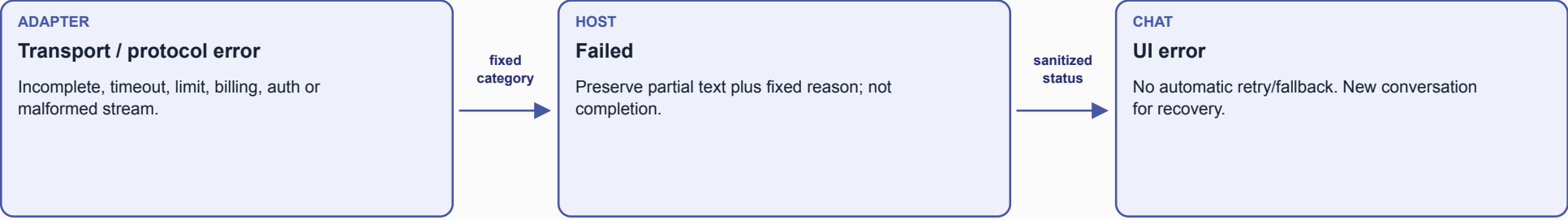
Evidence: HOST, CHAT

HOST · active cancellation



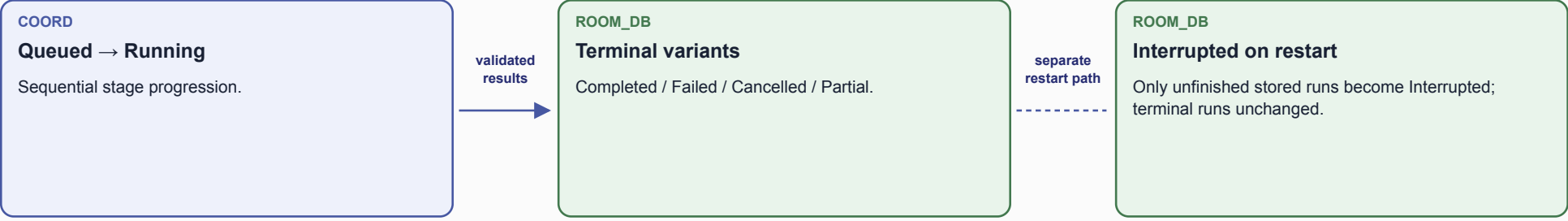
Evidence: CHAT, HOST, UI

HOST · independent failure



Evidence: ADAPTER, API, HOST, CHAT

ROOM_DB · durable collaboration states



Evidence: ROUTES, COORD, ROOM_DB

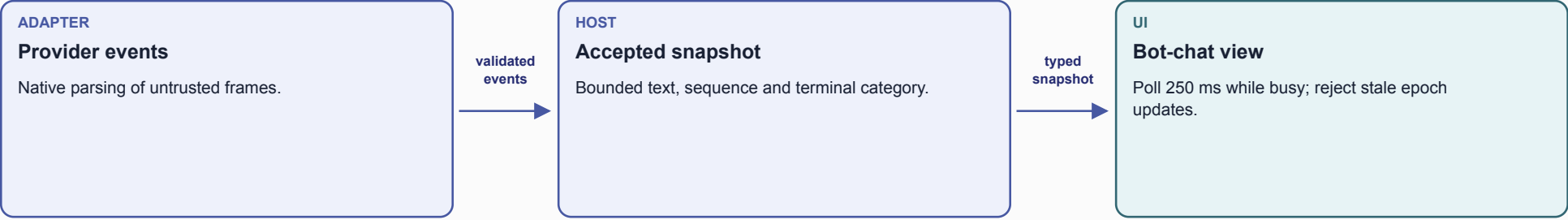
Awaiting approval belongs to synthetic/foundation workflows, not this live machine. Journal cap128 and text/token/time limits remain. Local Stop cannot prove zero remote billing or server cancellation.

06 · Observability and graph updates

Observable state and outputs; never private model thoughts

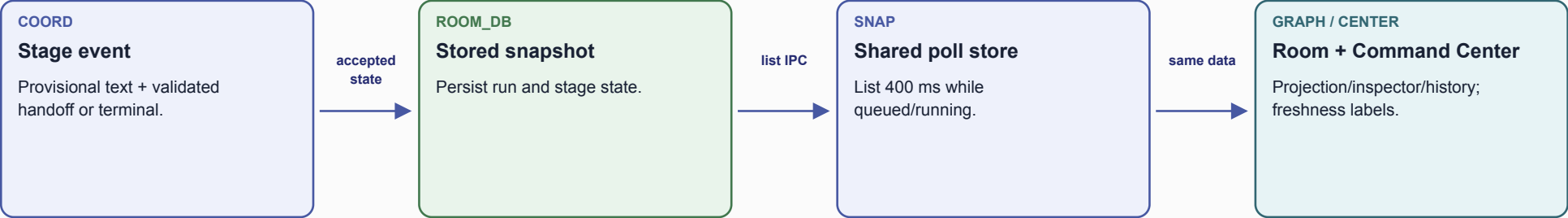
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Bot chat · polling, not a WebView provider subscription



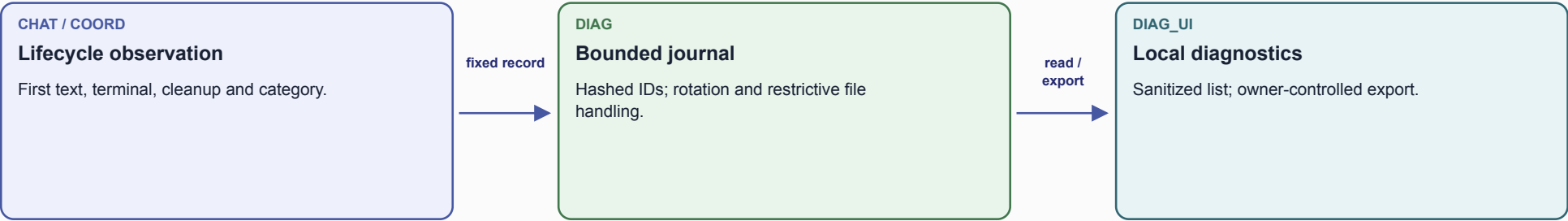
Evidence: ADAPTER, HOST, UI

Collaboration · one source for room and graph



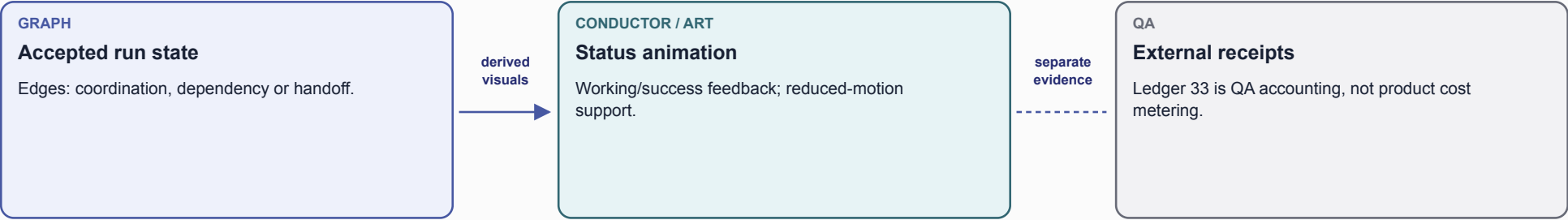
Evidence: COORD, ROOM_DB, SNAP, GRAPH, CENTER

Content-free operational diagnostics



Evidence: DIAG, DIAG_IPC, DIAG_UI

Presentation and distinct QA evidence



Evidence: GRAPH, CONDUCTOR, ART, REVIEW

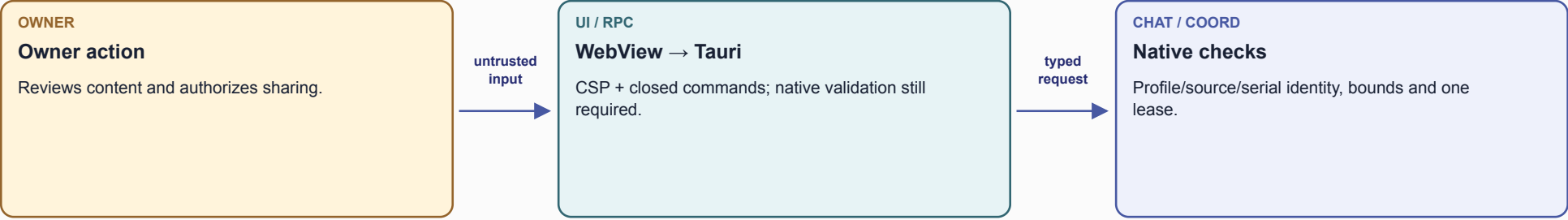
Mock tool events and deterministic demo graphs are separate. Diagnostics are not immutable audit, raw provider logs or a billing ledger. No generic live tool results feed this graph.

07 · Security and outbound data

Native typed boundaries; model text and WebView input remain untrusted

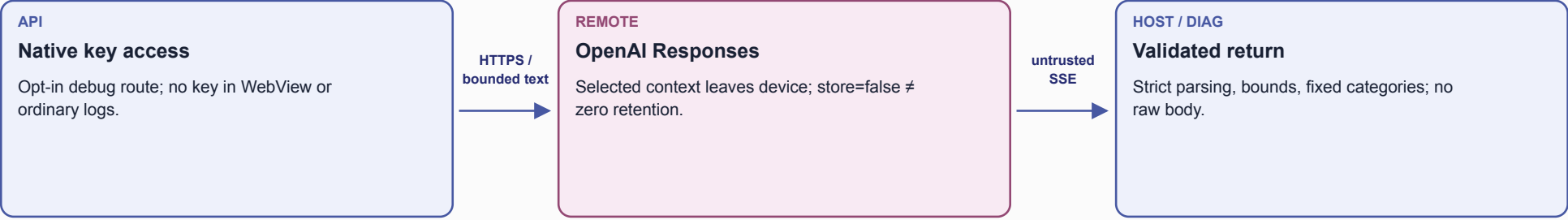
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Boundary A · WebView to native admission



Evidence: RPC, CSP, CAP, CHAT, COORD

Boundary B · direct HTTPS route



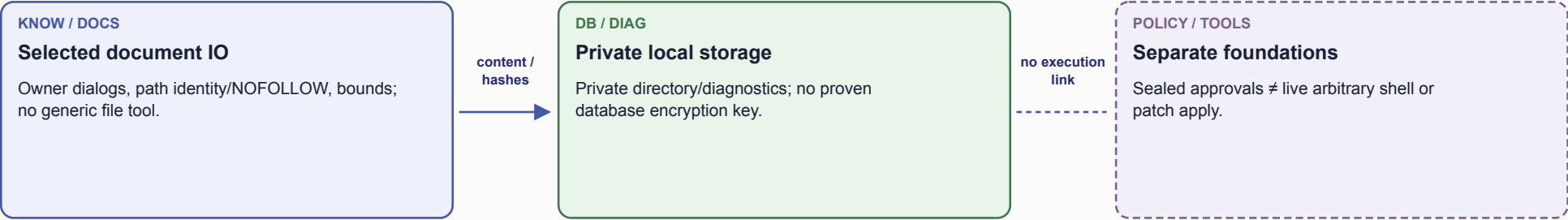
Evidence: API, HOST, DIAG

Boundary C · child runtime route



Evidence: CODEX, ADAPTER

Boundary D · local content and files



Evidence: DOCS, KNOW, DB, DIAG, POLICY, TOOLS

Roles/prompts are not enforced controls. A directory is not a sandbox. Runtime flags are not a full isolation audit. Prompt injection can influence text; schema validation does not establish truth.

08 · Development and delivery

Development pipeline is separate from native request execution

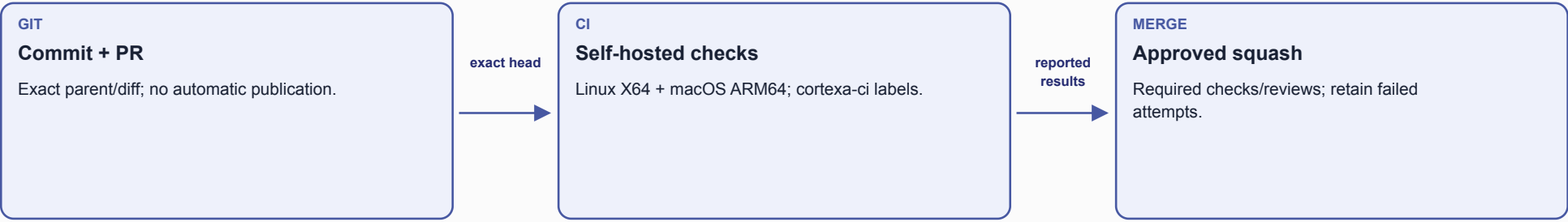
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Owner-approved local increment



Evidence: NPM, REVIEW

Publication · separately authorized



Evidence: CI, DOC_CI

Build and artifact provenance



Evidence: NPM, DEPS, REVIEW

Private development boundary



Evidence: REVIEW, UI_REVIEW, CI

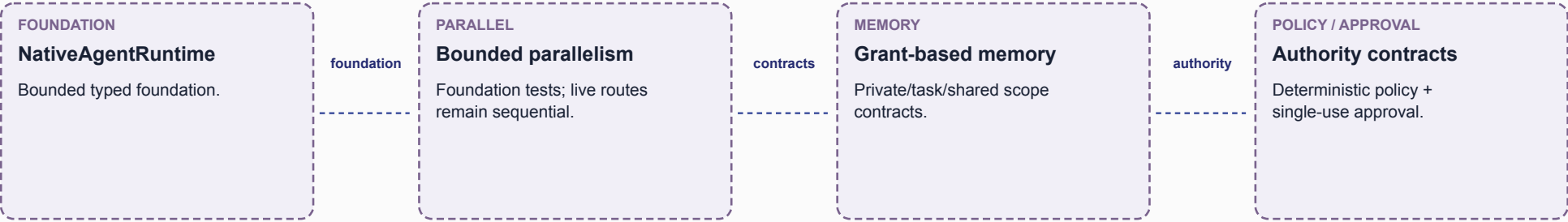
CI content permissions are read-only; ordinary jobs make no live requests. Self-hosted administration is an external trust boundary. No product tests/builds/live requests repeated for these diagrams.

09 · Future and separate foundations

Dashed nodes are not current live integration claims

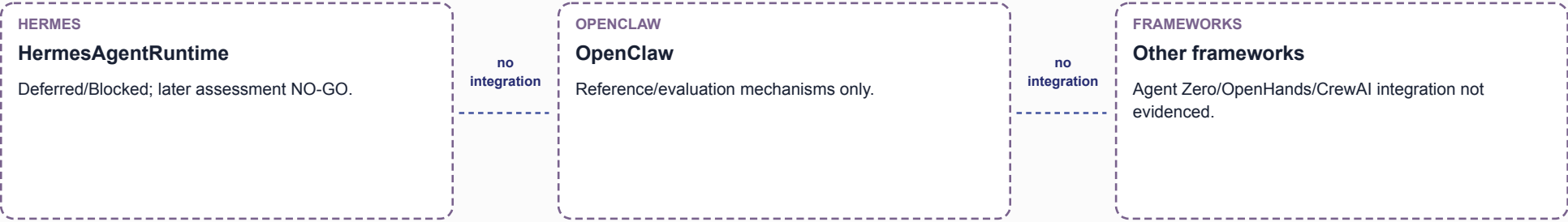
Native Interface Owner Remote Stored Separate foundation Deferred / unverified

Tested foundations · not a general live executor



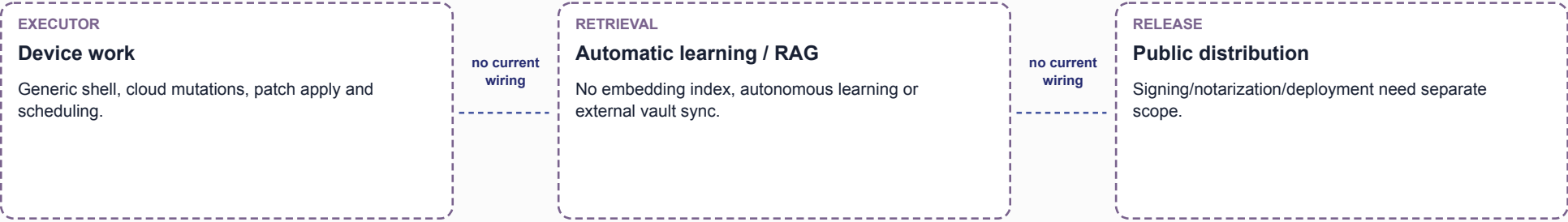
Evidence: FOUNDATION, PARALLEL, MEMORY, APPROVAL

External runtime direction · evaluation only



Evidence: HERMES, NATIVE_PLAN, PLAN

Not established in current application



Evidence: RPC, TOOLS, KNOW_DB, CI

Dashed connectors compare separate capabilities or directions; they are not runtime dataflow. D-125/M1/M2 stay parked. This document grants no future admission or implementation authority.